ic_autolab round 26 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T08:01:30Z at git `49addad3c9f544f7f7d89a80e47b0095ee8704c9`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-f9ed8c` (`python3 tools/pcc/pcc.py standing PCC-20261104-f9ed8c`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
28	**regression**	wsum2-b50-n2	1.066 [0.990, 1.148]	1.100 [1.015, 1.191]	1.263 [1.138, 1.402]	inc < sel	2.750 / 3.076 / 6.012	2.580 / 2.798 / 4.759	0.996 / 1.090 / 1.928	2.8x / 2.8x / 3.1x
32	**no_challenger**	none (best candidate `wsum2-b100-n2`, train S 2.682 / 3.030 / 6.161)	-	-	-	-	-	2.581 / 2.833 / 5.100	1.030 / 1.131 / 2.044	inc: 2.5x / 2.5x / 2.5x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 28: compare ops.status = ok, admissible = true; sel / inc = 1.0660 [0.990, 1.148] at w = 0; band: inc < sel; A/A inc-aa / inc interval [1.000, 1.000] contains 1
- k = 32: no_challenger

Train (development screen, in sample)

k = 28: session `ECBS1h05f96577a44c` (run exit 0, verify exit 0, receipt sha256 `70c59cc3e3652cb4cd280a8edae2d759aebce91a52f21b17ae02ca278bc2ad21`)

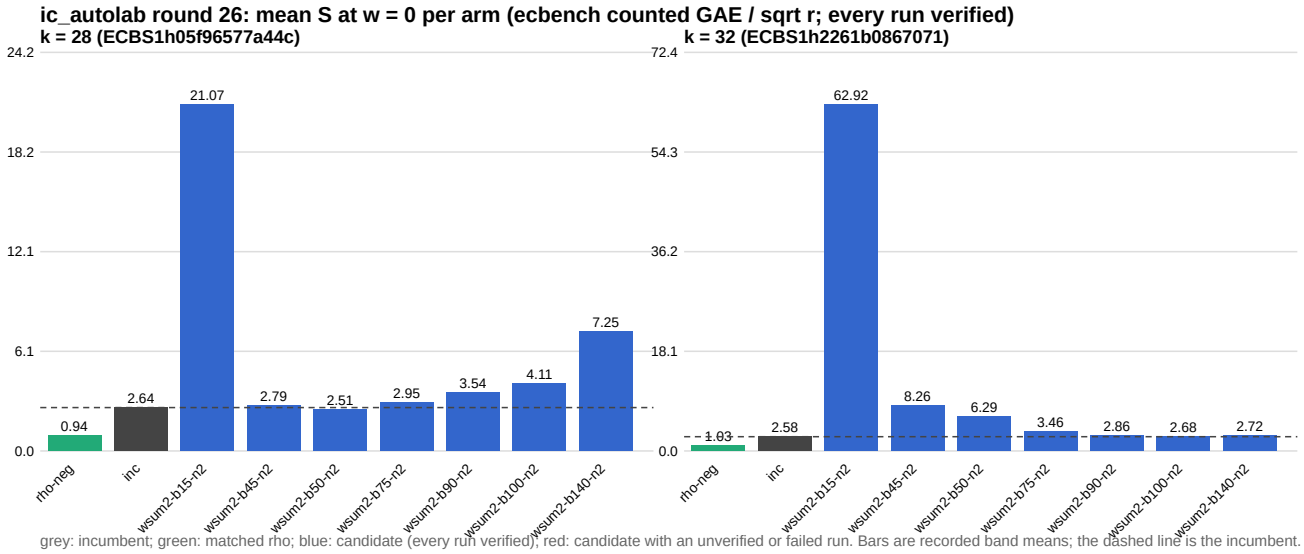
arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	`--`	`rho.negation`	32	32	0.944 / 1.032 / 1.824	matched reference
`inc`	wsum2-b60	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=60,free=1`	32	32	2.640 / 2.870 / 4.939	incumbent
`wsum2-b15-n2`	wsum2-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	21.067 / 25.177 / 62.166	cross-size transfer: wsum2-b15-n2 is the k = 20 incumbent at N / N_a = 0.062
`wsum2-b45-n2`	wsum2-b45	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=45,free=1`	32	32	2.790 / 3.157 / 6.458	exploration: family wsum2-b45 untried at k = 28, N = 0.0001 x N_a
`wsum2-b50-n2`	wsum2-b50	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=50,free=1`	32	32	2.505 / 2.784 / 5.290	exploration: family wsum2-b50 untried at k = 28, N = 0.0001 x N_a

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`wsum2-b75-n2`	wsum2-b75	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=75,free=1`	32	32	2.952 / 3.104 / 4.474	exploration: family wsum2-b75 untried at k = 28, N = 0.0001 x N_a
`wsum2-b90-n2`	wsum2-b90	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=90,free=1`	32	32	3.544 / 3.643 / 4.533	exploration: family wsum2-b90 untried at k = 28, N = 0.0001 x N_a
`wsum2-b100-n2`	wsum2-b100	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=100,free=1`	32	32	4.110 / 4.191 / 4.915	exploration: family wsum2-b100 untried at k = 28, N = 0.0001 x N_a
`wsum2-b140-n2`	wsum2-b140	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=140,free=1`	32	32	7.248 / 7.291 / 7.674	exploration: family wsum2-b140 untried at k = 28, N = 0.0001 x N_a

k = 32: session `ECBS1h2261b0867071` (run exit 0, verify exit 0, receipt sha256 `fd165eae09ca2d2b0f2353b81d8cb45dbac1c4cd4e3b4088efdd9b4f679298db`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.030 / 1.131 / 2.044	matched reference
`inc`	wsum2-b120	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	2.581 / 2.833 / 5.100	incumbent
`wsum2-b15-n2`	wsum2-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	62.921 / 75.466 / 188.366	cross-size transfer: wsum2-b15-n2 is the k = 20 incumbent at N / N_a = 0.062
`wsum2-b45-n2`	wsum2-b45	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=45,free=1`	32	32	8.260 / 9.854 / 24.203	exploration: family wsum2-b45 untried at k = 32, N = 0.0001 x N_a
`wsum2-b50-n2`	wsum2-b50	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=50,free=1`	32	32	6.293 / 7.487 / 18.230	exploration: family wsum2-b50 untried at k = 32, N = 0.0001 x N_a
`wsum2-b75-n2`	wsum2-b75	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=75,free=1`	32	32	3.462 / 4.040 / 9.243	exploration: family wsum2-b75 untried at k = 32, N = 0.0001 x N_a
`wsum2-b90-n2`	wsum2-b90	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=90,free=1`	32	32	2.863 / 3.280 / 7.029	exploration: family wsum2-b90 untried at k =

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
wsum2-b100-n2	wsum2-b100	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=100,free=1	32	32	2.682 / 3.030 / 6.161	exploration: family wsum2-b100 untried at k = 32, N = 0.0001 x N_a
wsum2-b140-n2	wsum2-b140	2 prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=140,free=1	32	32	2.721 / 2.911 / 4.622	exploration: family wsum2-b140 untried at k = 32, N = 0.0001 x N_a



train screen

Selection

- k = 28: challenger `wsum2-b50-n2` (Pareto alternative at w = 1: `wsum2-b75-n2`)
- k = 32: no_challenger (Pareto alternative at w = 1: `wsum2-b140-n2`)

Holdout sessions

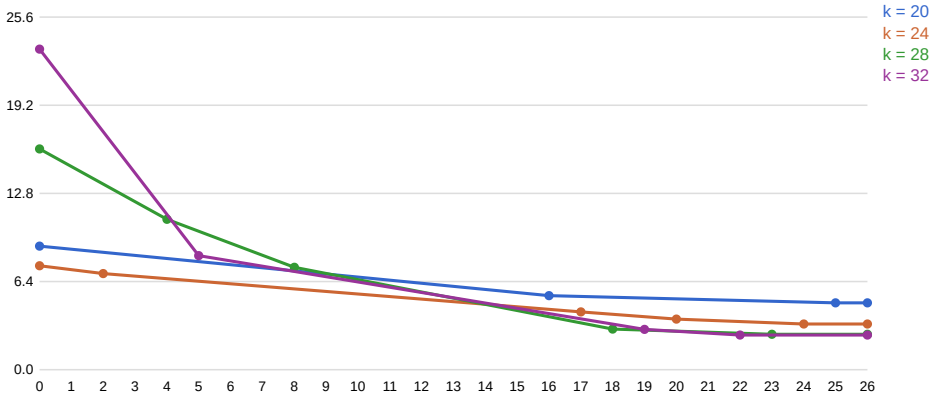
k	session	run exit	verify exit	receipt sha256	levels
28	ECBS1he23f5c757de8	0	0	29ac5ddc611d0ff6addb5c2c674860aee1438fa986a0f1426c63dc16c6c03f4c	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	wsum2-b15-n2 (prime-projected-abscissa:size=2,scan_cap=32 + wsum:coef=15,free=1)	25	4.860 / 5.083 / 7.092	3.2x / 3.2x / 2.8x
24	wsum2-b30-n2 (prime-projected-abscissa:size=2,scan_cap=32 + wsum:coef=30,free=1)	24	3.318 / 3.573 / 5.863	2.7x / 2.7x / 2.5x
28	wsum2-b60-n2 (prime-projected-abscissa:size=2,scan_cap=32 + wsum:coef=60,free=1)	23	2.576 / 2.793 / 4.746	2.8x / 2.7x / 2.6x

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
32	`wsum2-b120-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=120,free=1`)	22	2.519 / 2.759 / 4.915	2.7x / 2.7x / 2.7x

Ratchet: incumbent mean S at w = 0 by round (generic track, Curve25519 miniatures)



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.